

Linux Commands:

```
ubuntu@ip-10-0-2-155:~$ whoami
ubuntu
ubuntu@ip-10-0-2-155:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,,:/var/lib/tpm:/bin/false
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:106:1::/var/cache/pollinate:/bin/false
tcpdump:x:107:108::/nonexistent:/usr/sbin/nologin
```

```
ubuntu@ip-10-0-2-155:~$ ls
commands.txt  devops
ubuntu@ip-10-0-2-155:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-2-155:~$ cd /home
ubuntu@ip-10-0-2-155:/home$ ls
babijati  bhide  ubuntu
ubuntu@ip-10-0-2-155:/home$ sudo userdel bijati
userdel: user 'bijati' does not exist
ubuntu@ip-10-0-2-155:/home$ sudo useradd -m babitaji
ubuntu@ip-10-0-2-155:/home$ ls
babijati  babitaji  bhide  ubuntu
ubuntu@ip-10-0-2-155:/home$ sudo passwd babitaji
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-10-0-2-155:/home$ pwd
/home
```

```
ubuntu@ip-10-0-2-155:/home$ whoami
ubuntu
ubuntu@ip-10-0-2-155:/home$ sudo su babitaji/
su: user babitaji/ does not exist or the user entry does not contain all the required fields
ubuntu@ip-10-0-2-155:/home$ su babitaji/
su: user babitaji/ does not exist or the user entry does not contain all the required fields
ubuntu@ip-10-0-2-155:/home$ su babitaji
Password:
$ whoami
babitaji
```

```
ubuntu@ip-10-0-2-155:/home$ sudo useradd -m tappu -s /bin/bash
ubuntu@ip-10-0-2-155:/home$ ls
babijati babitaji bhide tappu ubuntu
ubuntu@ip-10-0-2-155:/home$ sudo passwd tappu
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-10-0-2-155:/home$ su tappu
Password:
tappu@ip-10-0-2-155:/home$ whoami
tappu
tappu@ip-10-0-2-155:/home$ echo $SHELL
/bin/bash
tappu@ip-10-0-2-155:/home$
```

```
ubuntu@ip-10-0-2-155:~$ echo "Hello Dastoo.."
Hello Dastoo..
ubuntu@ip-10-0-2-155:~$ ls
commands.txt devops
ubuntu@ip-10-0-2-155:~$ rm commands.txt
ubuntu@ip-10-0-2-155:~$ ls
devops
ubuntu@ip-10-0-2-155:~$ rm -r devops/
ubuntu@ip-10-0-2-155:~$ ls
ubuntu@ip-10-0-2-155:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-2-155:~$ mkdir devops
ubuntu@ip-10-0-2-155:~$ ls
devops
ubuntu@ip-10-0-2-155:~$ touch myfile.txt
ubuntu@ip-10-0-2-155:~$ ls
devops myfile.txt
ubuntu@ip-10-0-2-155:~$ cd devops
ubuntu@ip-10-0-2-155:~/devops$ touch intro.txt
ubuntu@ip-10-0-2-155:~/devops$ ls
intro.txt
ubuntu@ip-10-0-2-155:~/devops$ pwd
/home/ubuntu/devops
ubuntu@ip-10-0-2-155:~/devops$ cd ..
ubuntu@ip-10-0-2-155:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-2-155:~$ cd ..
ubuntu@ip-10-0-2-155:/home$ pwd
/home
ubuntu@ip-10-0-2-155:/home$ cd ..
ubuntu@ip-10-0-2-155:/$
```

```
ubuntu@ip-10-0-2-155:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.7).
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
ubuntu@ip-10-0-2-155:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-05 10:47:27 UTC; 9min ago
     Docs: man:nginx(8)
  Main PID: 3582 (nginx)
    Tasks: 3 (limit: 1013)
  Memory: 2.6M (peak: 5.3M)
    CPU: 31ms
   CGroup: /system.slice/nginx.service
           └─3582 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─3584 "nginx: worker process"
               └─3585 "nginx: worker process"

May 05 10:47:27 ip-10-0-2-155 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 05 10:47:27 ip-10-0-2-155 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-10-0-2-155:~$
```

```
ubuntu@ip-10-0-2-155:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,ubuntu
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
```

```
ubuntu@ip-10-0-2-155:~$ sudo groupadd devops
ubuntu@ip-10-0-2-155:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,ubuntu
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:

```

```
ubuntu@ip-10-0-2-155:~$ sudo gpasswd -M babijati tester
ubuntu@ip-10-0-2-155:~$ cat /etc/group
```

```

ubuntu@ip-10-0-2-155:~/logs$ grep -i "session open" app.log
Jun 15 04:06:18 combo su(pam_unix)[21416]: session opened for user cyrus by (uid=0)
Jun 15 04:12:42 combo su(pam_unix)[22644]: session opened for user news by (uid=0)
Jun 16 04:10:22 combo su(pam_unix)[25178]: session opened for user cyrus by (uid=0)
Jun 16 04:16:17 combo su(pam_unix)[25548]: session opened for user news by (uid=0)
Jun 17 04:03:33 combo su(pam_unix)[27953]: session opened for user cyrus by (uid=0)
Jun 17 04:09:43 combo su(pam_unix)[29190]: session opened for user news by (uid=0)
Jun 17 20:29:26 combo sshd(pam_unix)[30631]: session opened for user test by (uid=509)
Jun 18 04:07:05 combo su(pam_unix)[31791]: session opened for user cyrus by (uid=0)
Jun 18 04:12:42 combo su(pam_unix)[32164]: session opened for user news by (uid=0)
Jun 19 04:08:55 combo su(pam_unix)[2192]: session opened for user cyrus by (uid=0)
Jun 19 04:15:18 combo su(pam_unix)[3676]: session opened for user news by (uid=0)
Jun 20 04:02:54 combo su(pam_unix)[9187]: session opened for user cyrus by (uid=0)
Jun 20 04:08:37 combo su(pam_unix)[9558]: session opened for user news by (uid=0)
Jun 21 04:06:57 combo su(pam_unix)[12098]: session opened for user cyrus by (uid=0)
Jun 21 04:13:03 combo su(pam_unix)[13327]: session opened for user news by (uid=0)
Jun 22 04:05:58 combo su(pam_unix)[16663]: session opened for user cyrus by (uid=0)
Jun 22 04:11:42 combo su(pam_unix)[17037]: session opened for user news by (uid=0)
Jun 23 04:05:28 combo su(pam_unix)[19534]: session opened for user cyrus by (uid=0)
Jun 23 04:13:08 combo su(pam_unix)[24180]: session opened for user news by (uid=0)
Jun 24 04:05:34 combo su(pam_unix)[26938]: session opened for user cyrus by (uid=0)
Jun 24 04:11:12 combo su(pam_unix)[27311]: session opened for user news by (uid=0)
Jun 25 04:04:25 combo su(pam_unix)[29690]: session opened for user cyrus by (uid=0)
Jun 25 04:10:34 combo su(pam_unix)[30934]: session opened for user news by (uid=0)
Jun 26 04:04:17 combo su(pam_unix)[945]: session opened for user cyrus by (uid=0)
Jun 26 04:10:02 combo su(pam_unix)[1546]: session opened for user news by (uid=0)
Jun 27 04:02:47 combo su(pam_unix)[7031]: session opened for user cyrus by (uid=0)
Jun 27 04:08:56 combo su(pam_unix)[8269]: session opened for user news by (uid=0)
Jun 28 04:03:15 combo su(pam_unix)[10735]: session opened for user cyrus by (uid=0)
Jun 28 04:09:00 combo su(pam_unix)[11106]: session opened for user news by (uid=0)
Jun 29 04:03:10 combo su(pam_unix)[13665]: session opened for user cyrus by (uid=0)
Jun 29 04:09:29 combo su(pam_unix)[14891]: session opened for user news by (uid=0)
Jun 30 04:03:41 combo su(pam_unix)[17407]: session opened for user cyrus by (uid=0)
Jun 30 04:09:30 combo su(pam_unix)[17778]: session opened for user news by (uid=0)
Jun 30 22:16:32 combo sshd(pam_unix)[19432]: session opened for user test by (uid=509)
Jun 30 22:16:32 combo sshd(pam_unix)[19431]: session opened for user test by (uid=509)
Jun 30 22:16:32 combo sshd(pam_unix)[19433]: session opened for user test by (uid=509)

```

```

ubuntu@ip-10-0-2-155:~/logs$ grep -i "Authentication Failure" app.log | tail -n 5
Jul 26 07:03:57 combo sshd(pam_unix)[28878]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114
user=root
Jul 26 07:04:02 combo sshd(pam_unix)[28880]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114
user=root
Jul 26 07:04:05 combo sshd(pam_unix)[28882]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114
user=root
Jul 26 07:04:07 combo sshd(pam_unix)[28884]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114
user=root
Jul 26 07:04:12 combo sshd(pam_unix)[28886]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114
user=root
ubuntu@ip-10-0-2-155:~/logs$

```

```

ubuntu@ip-10-0-2-155:~/logs$ awk '/authentication failure/ {print $12,$13,$14,$15,$16}' app.log | head
ruser= rhost=218.188.2.4
ruser= rhost=218.188.2.4
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
ubuntu@ip-10-0-2-155:~/logs$ awk '/authentication failure/ {print $12,$13,$14,$15,$16}' app.log | tail -n 5
ruser= rhost=207.243.167.114 user=root
ruser= rhost=207.243.167.114 user=root
ruser= rhost=207.243.167.114 user=root
ruser= rhost=207.243.167.114 user=root
ruser= rhost=207.243.167.114 user=root
ubuntu@ip-10-0-2-155:~/logs$

```

```

ubuntu@ip-10-0-2-155:~/logs$ awk '/authentication failure/ { if($1=="Jul" && ($2=="11")) print $1,$2,$12,$13,$14,$15,$16}' app.log
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=82.77.200.128 user=root
Jul 11 ruser= rhost=
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
Jul 11 ruser= rhost=211.137.205.253 user=root
ubuntu@ip-10-0-2-155:~/logs$ awk '/authentication failure/ { if($1=="Jul" && ($2=="11") && ($3=="17:58:16")) print $1,$2,$3,$12,$13,$14,$15,$16}' app.log
Jul 11 17:58:16 ruser= rhost=211.137.205.253 user=root
Jul 11 17:58:16 ruser= rhost=211.137.205.253 user=root
Jul 11 17:58:16 ruser= rhost=211.137.205.253 user=root
ubuntu@ip-10-0-2-155:~/logs$

```

```

root@ip-10-0-2-70:/$ mkdir /mnt/tws-disk-mount
root@ip-10-0-2-70:/$ cd /mnt
root@ip-10-0-2-70:/mnt$ ls
tws-disk-mount  tws-lv-mount
root@ip-10-0-2-70:/mnt$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   74M  1 loop /snap/core22/2411
loop1                               7:1      0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop2                               7:2      0  48.4M  1 loop /snap/snapd/26382
nvme0n1                             259:0    0    8G   0 disk
├─nvme0n1p1                         259:4    0   6.9G  0 part /
├─nvme0n1p13                        259:5    0 1023M  0 part /boot
├─nvme0n1p14                        259:6    0    4M   0 part
└─nvme0n1p15                        259:7    0 106M   0 part /boot/efi
nvme1n1                             259:1    0   12G  0 disk
└─tws--vg-tws--lv                  252:0    0   10G  0 lvm
nvme2n1                             259:2    0   14G  0 disk
nvme3n1                             259:3    0   10G  0 disk
root@ip-10-0-2-70:/mnt$ mkfs -t ext4 /dev/nvme3n1
mke2fs 1.47.2 (1-Jan-2025)
/dev/nvme3n1 contains a LVM2_member file system
Proceed anyway? (y,N) y
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: 7cd08741-7b49-4c43-9316-3681bae51518
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

```

```

ubuntu@ip-10-0-2-155:~$ mkdir logs
ubuntu@ip-10-0-2-155:~$ cd logs/
ubuntu@ip-10-0-2-155:~/logs$ ls
ubuntu@ip-10-0-2-155:~/logs$ cd ..
ubuntu@ip-10-0-2-155:~$ ls
commands.txt  demo.txt  devops  logs
ubuntu@ip-10-0-2-155:~$ cd /var
ubuntu@ip-10-0-2-155:/var$ ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
ubuntu@ip-10-0-2-155:/var$ cd log
ubuntu@ip-10-0-2-155:/var/log$ ls
README      auth.log      cloud-init-output.log  dmesg.1.gz  journal      nginx      ubuntu-advantage-apt-hook.log
alternatives.log  auth.log.1    cloud-init.log         dmesg.2.gz  kern.log     private   unattended-upgrades
alternatives.log.1  auth.log.2.gz cloud-init.log.1       dmesg.3.gz  kern.log.1   syslog    wtmp
amazon         btmap        dist-upgrade          dmesg.4.gz  kern.log.2.gz syslog.1
apport.log      btmap.1      dmesg                 dpkg.log    landscape    syslog.2.gz
apt            chrony       dmesg.0              dpkg.log.1  lastlog      sysstat
ubuntu@ip-10-0-2-155:/var/log$ cd nginx
ubuntu@ip-10-0-2-155:/var/log/nginx$ ls
access.log  access.log.2.gz  access.log.4.gz  access.log.6.gz  access.log.8.gz  error.log.1
access.log.1  access.log.3.gz  access.log.5.gz  access.log.7.gz  error.log        error.log.2.gz
ubuntu@ip-10-0-2-155:/var/log/nginx$ cat access.log
204.76.203.206 - - [10/May/2026:08:10:26 +0000] "GET / HTTP/1.1" 200 19848 "-" "Mozilla/5.0"
54.239.99.148 - - [10/May/2026:08:13:38 +0000] "GET /__aes/proxy_health HTTP/1.1" 404 134 "-" "CloudSearch/ControlPlane/CheckLimits"
87.236.176.244 - - [10/May/2026:08:15:09 +0000] "GET / HTTP/1.1" 200 6639 "-" "Mozilla/5.0 (compatible; InternetMeasurement/1.0; +http://internet-measurement.com/)"
204.76.203.206 - - [10/May/2026:08:20:08 +0000] "GET / HTTP/1.1" 200 19848 "-" "Mozilla/5.0"
ubuntu@ip-10-0-2-155:/var/log/nginx$

```

```

root@ip-10-0-2-9:/$ lvm
lvm> pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1
Physical volume "/dev/nvme1n1" successfully created.
Physical volume "/dev/nvme2n1" successfully created.
Physical volume "/dev/nvme3n1" successfully created.
lvm> pvs
PV          VG Fmt Attr PSize  PFree
/dev/nvme1n1    lvm2 ---  10.00g 10.00g
/dev/nvme2n1    lvm2 ---  12.00g 12.00g
/dev/nvme3n1    lvm2 ---  14.00g 14.00g
lvm> vgcreate tws-vg /dev/nvme1n1 /nvme2n1
No device found for /nvme2n1.
Command requires all devices to be found.
lvm> vgcreate tws-vg /dev/nvme1n1 /dev/nvme2n1
Volume group "tws-vg" successfully created
lvm> vgs
VG      #PV #LV #SN Attr   VSize  VFree
tws-vg   2   0   0 wz--n- 21.99g 21.99g
lvm> lvcreate -L 10G -n tws-lv tws-vg
Logical volume "tws-lv" created.
lvm> lvs
LV      VG      Attr      LSize  Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
tws-lv tws-vg   -wi-a----- 10.00g

```

```

root@ip-10-0-2-9:/$ lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0             7:0    0   74M  1 loop /snap/core22/2411
loop1             7:1    0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop2            7:2    0  48.4M  1 loop /snap/snapd/26382
nvme0n1          259:0    0    8G   0 disk
├─nvme0n1p1       259:1    0   6.9G   0 part /
├─nvme0n1p13      259:2    0 1023M   0 part /boot
├─nvme0n1p14      259:3    0    4M   0 part
└─nvme0n1p15      259:4    0 106M   0 part /boot/efi
nvme1n1          259:5    0   10G   0 disk
nvme2n1          259:6    0   12G   0 disk
└─tws--vg-tws--lv 252:0    0   10G   0 lvm
nvme3n1          259:7    0   14G   0 disk
root@ip-10-0-2-9:/$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  31% /
tmpfs           455M    0  455M   0% /dev/shm
tmpfs           182M  928K  181M   1% /run
efivarfs        128K   3.1K  120K   3% /sys/firmware/efi/efivars
none            1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           455M    0  455M   0% /tmp
none            1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  826M  11% /boot
/dev/nvme0n1p15  105M   6.3M   99M   7% /boot/efi
none            1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
none            1.0M    0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           91M   8.0K   91M   1% /run/user/1000
root@ip-10-0-2-9:/$

```